

CHAPTER 23

REHABILITATION, RECONDITIONING, AND MEDICAL ISSUES

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After completing this chapter, you will be able to

- identify the members of the allied health team and their responsibilities during the rehabilitation and reconditioning of injured athletes;
- recognize the types of injuries and medical conditions that athletes sustain;
- comprehend the timing and events of tissue healing;
- understand the goals of each tissue healing phase; and
- describe the role of the strength and conditioning professional during injury rehabilitation and reconditioning.

As the employment of strength and conditioning professionals continues to increase, their role will expand to allow more active

participation in the rehabilitation and reconditioning of injured athletes. The strength and conditioning professional has unique knowledge and insight regarding optimal athletic function and can serve a vital role during the final stages of an advanced rehabilitation program by preparing the athlete for a return to competition. These abilities give the strength and conditioning professional distinct responsibilities during rehabilitation from athletic injury. To fully understand how the strength and conditioning professional can best augment the rehabilitation of injured athletes, one must first recognize the role of each member of the allied health team. Furthermore, an understanding of different types of injury and the physiological healing process is essential to hastening the recovery from injury.

This chapter is not intended to provide the reader with rehabilitation protocols for specific injuries. Rather, the aim is to explain the physiological events that follow a musculoskeletal injury, thereby allowing optimal goal setting by the strength and conditioning professional to improve rehabilitation outcomes. The information in this chapter should ultimately be used to return an injured athlete to maximal function. The rehabilitation and reconditioning approach follows five basic principles, listed in the sidebar.

Allied Health Team

The **allied health** team provides health care services, with athlete needs and concerns as the primary focus (15). All members of the allied health team are responsible for educating coaches and athletes regarding injury risks, precautions, and treatments; they must also prevent injuries and rehabilitate those athletes who have sustained injury. Several different professionals play important roles in assisting the injured athlete's return to the playing field, necessitating effective communication.

Principles of Rehabilitation and Reconditioning

- Healing tissues must not be overstressed.
- The athlete must fulfill specific criteria to progress from one phase to another during the rehabilitative process.
- The rehabilitation program must be based on current clinical and scientific research.
- The program must be adaptable to each individual and his or her specific requirements and goals.
- Rehabilitation is a team-oriented process requiring all the members of the allied health team to work together toward a common goal of returning the athlete to unrestricted competition as quickly and safely as possible.

Allied Health Team Members

The **team physician** provides medical care to an organization, school, or team. The team physician is most often a medical doctor (MD) or doctor of osteopathy (DO). The team physician may have specialized training (residency or fellowship) in a variety of fields, including family medicine, internal medicine, pediatrics, and orthopedics (67) but should be proficient in the care of musculoskeletal injuries and sport-related medical conditions (32). Specific responsibilities of the team physician may include

preparticipation examinations, on-field emergency care, injury and illness evaluation and diagnosis, and referral to other health care professionals as needed (67). Although not responsible for daily rehabilitation, the team physician most often makes the final determination of an athlete's readiness for return to competition (31, 67). Another important role of the team physician is the prescription of medications as needed, including anti-inflammatory, pain, and cold and flu medications.

The person typically responsible for the day-to-day physical health of the athlete is the **athletic trainer** or athletic therapist. In the United States, an athletic trainer is certified by the National Athletic Trainers' Association Board of Certification as a Certified Athletic Trainer (ATC). The athletic trainer works under the supervision of the team physician and is employed primarily by secondary schools, colleges, or professional teams but may also work in an outpatient physical therapy clinic. Primary responsibilities of this individual include management and rehabilitation of injuries resulting from physical activity and prevention of injuries through the prescription of sport-specific exercise and the application of prophylactic equipment (e.g., tape and braces). Specifically, the athletic trainer evaluates injuries, provides the injured athlete with therapeutic exercise to hasten the rehabilitation process, treats injuries with therapeutic modalities, and serves as an administrator for the allied health team (15). Because the athletic trainer has a significant amount of contact with the athlete, he or she plays a key role in promoting communication between members of the allied health team, the coach, and the athlete (15).

A **physical therapist** (or **physiotherapist**) with specialization in orthopedics or sport physical therapy can play a valuable role in reducing pain and restoring function to the injured athlete. Although physical therapists are typically based in outpatient physical therapy clinics, many collegiate and professional teams now directly employ physical therapists as members of the allied health staff. A physical therapist may help develop specific treatment strategies or manage

long-term rehabilitation. In the United States, physical therapists with expertise in sport injury management or orthopedics may become board certified through the American Board of Physical Therapy Specialties (ABPTS) and obtain the Sports Certified Specialist (SCS) or Orthopedic Certified Specialist (OCS) credentials, respectively. These board-certified specialists are more commonly becoming participants in the evaluation, treatment, and rehabilitation of acutely injured athletes and often serve in the dual capacity of team athletic trainer and physical therapist.

The **strength and conditioning professional** typically focuses on strength, power, and performance enhancement. The strength and conditioning professional also plays a valuable role within the allied health team and is an integral part of the rehabilitation and reconditioning process. Ideally, this person should be certified by the National Strength and Conditioning Association as a Certified Strength and Conditioning Specialist (CSCS) to ensure that he or she has the knowledge and background to contribute to the rehabilitation process. In consultation with the athletic trainer or physical therapist, this professional uses an understanding of the proper technique and application of several types of exercise (e.g., resistance, plyometric, and aerobic exercise) to develop a reconditioning program to ready the injured athlete for return to competition. Furthermore, strength and conditioning professionals possess an extensive understanding of the role that biomechanics play in a wide variety of sports and activities, which may allow them to suggest exercises for advanced rehabilitation and reconditioning of many injuries.

Additionally, the allied health team often includes specialized members who assist with the postacute rehabilitation and reconditioning of injured athletes. An **exercise physiologist** has a formal background in the exercise sciences and uses his or her expertise to assist with the design of a conditioning program that carefully considers the body's metabolic response to exercise and the ways in which that reaction aids the healing process. Because proper

nutrition is crucial in recovery from injury, a **nutritionist** or a registered dietitian with a background in sport nutrition may provide guidelines regarding proper food choices to optimize tissue recovery. Ideally, the nutritionist has been formally trained in food and nutrition sciences and may be a registered dietitian (RD) recognized by the Academy of Nutrition and Dietetics Commission on Dietetic Registration (15). A licensed **counselor**, **psychologist**, or **psychiatrist** with a background in sport may provide strategies that help the injured athlete better cope with the mental stress accompanying an injury.

Communication

Communication between members of the allied health team is essential (15, 97). Most often the injured athlete has the most contact with coaches, the athletic trainer, and the strength and conditioning professional. In some instances, athletes disclose the initial onset of an injury to a sport coach or strength and conditioning professional before consulting with the athletic trainer. Thus consistent communication between these individuals is imperative. This does not negate communication with other members of the allied health team (physician, physical therapist, nutritionist, sport psychologist) who may not have as frequent interactions over the course of a week. A weekly meeting of the allied health team can provide a forum so all members are able to discuss the training requirements and necessary restrictions for each injured athlete. Relevant questions for discussion could include the following: What is the current status of the athlete (no participation, limited participation, full participation)? What exercises or activities is the athlete currently performing? Are any restrictions or modifications necessary? How is the athlete progressing? Do any program changes need to be made?

To develop training programs most effectively for an injured athlete, the strength and conditioning professional must understand the diagnosis of the given injury and exercise indications and

contraindications. An **indication** is a form of treatment required by the rehabilitating athlete. For example, a softball outfielder with shoulder impingement must maintain lower extremity function, so the athletic trainer may request that the athlete continue to perform lower extremity strength, speed, agility, and power exercises during shoulder rehabilitation. Therefore, lower body exercise is indicated. A **contraindication** is an activity or practice that is inadvisable or prohibited due to the given injury. For example, during the later phases of rehabilitation from an anterior shoulder dislocation, an American football player may require upper body strengthening before being cleared to play. The athletic trainer requests that the athlete initiate upper body strengthening, but the bench press exercise may be contraindicated because it can place the injured shoulder in a vulnerable position in the presence of anterior instability. To clarify the role of the strength and conditioning professional during this process, it may be beneficial to have the allied health team use a form that specifies the indications and contraindications for exercises, providing for safe and efficient conditioning ([figure 23.1](#)). Additionally, the strength and conditioning professional can use a similar form ([figure 23.2](#)) to communicate the components of a given program and the athlete's subjective and objective responses to those components.

The allied health team includes a large number of professionals working together to provide an optimal rehabilitation and reconditioning environment. Thus, the relationship of the allied health team members requires thoughtful communication to ensure a safe, harmonious climate for the injured athlete.

Types of Injury

Macrotrauma is a specific, sudden episode of overload injury to a given tissue, resulting in disrupted tissue integrity. Trauma to the bone can lead to either a contusion or a fracture. Skeletal fractures can result from a direct blow to a bone and can be given a variety of classifications (e.g., closed, open, avulsed, incomplete). Joint trauma is manifested as either a **dislocation** (complete displacement of the joint surfaces) or a **subluxation** (partial displacement of the joint surfaces) and may result in joint laxity or instability. Ligamentous trauma is termed a **sprain** and is assigned a classification of *first degree* (partial tear of the ligament without increased joint instability), *second degree* (partial tear with minor joint instability), or *third degree* (complete tear with full joint instability).

Rehabilitation Referral

Date: January 2, 2026

Name: Allison Pierson

Sport and position: Volleyball setter

Injury date: November 21, 2025

Surgery date: December 4, 2025

Diagnosis: Left ACL reconstruction

Indications

Stationary bicycle: Progress gradually up to 60 minutes. No running at this time.

Single-leg hip sled: Less than 90 degrees of left knee flexion; begin without resistance; progress only after consultation with athletic training staff.

Upper extremity resistance exercises

Contraindications

Leg extension exercise

Full squats

Plyometrics

Running

Jonah Grey, ATC

January 2, 2026

FIGURE 23.1 A sample rehabilitation referral form, which communicates indications and contraindications to allied health team members.

Musculotendinous trauma is classified as either a **contusion** (if the trauma was direct) or a strain (if the trauma was indirect). A muscle contusion is an area of excess accumulation of blood and fluid in the tissues surrounding the injured muscle; it may severely limit the function of the injured muscle. Muscle **strains** are tears of muscle fibers and are further assigned grades, or degrees. A first-degree strain is a partial tear of individual fibers and is characterized by strong but painful muscle activity. A second-degree strain is a partial tear with weak, painful muscle activity. A third-degree muscle strain is a complete tear of the fibers and is manifested by very weak, painless muscle activity. A tendon, like a muscle, can also rupture if the tensile load applied to it exceeds its limit. Typically, tendon collagen fibers are significantly stronger than the muscle fibers to which they attach, so failure is more likely to occur within the muscle belly, at the musculotendinous junction, or at the tendon attachment to the bone than within the substance of the tendon.

Microtrauma, or overuse injury, results from repeated, abnormal stress applied to a tissue by continuous training or training with too little recovery time. Overuse injuries may be due to training errors (e.g., poor program design and excessive volume), suboptimal training surfaces (e.g., too hard or uneven), faulty biomechanics or technique during performance, insufficient motor control, decreased flexibility, or skeletal malalignment and predisposition (86, 97). Two common overuse injuries involve bone and tendon. The most common overuse injury to bone is a stress fracture. Although body type and structure, nutrition, and metabolic factors play a large role, stress fractures are often the result of a rapid increase in training volume or excessive training volume on hard training surfaces (4, 97). **Tendinitis** is inflammation of a tendon (83), and if the cause of the inflammation is left uncorrected, chronic tendinitis or tendinopathy may develop. (A term with the suffix *-itis* refers to an inflammatory condition [e.g., tendinitis and arthritis].) **Tendinopathy** is a degenerative condition characterized by minimal

inflammation, tendon cell damage or death, and **neovascularization** (83).

Strength and Conditioning Summary

Date: October 23, 2025

Name: Molly Jackson

Sport: Soccer

Position: Midfield

Diagnosis: Grade II right MCL sprain

Injury date: October 9, 2025

Activity Summary

Number of sessions: 7

Date begun: October 16, 2025

Current Activities

Activity	Sets	Reps	Resistance (lb)
Half squat	3	10	115
Leg extension	3	10	60
Leg flexion	3	10	50
Heel raise	3	15	95
Time		Speed	
Stationary bicycle	20 min		80 rpm
Stair stepper	20 min		70 ft per min
Jogging	10 min		5.0 mph

Assessment

No difficulty with half squats; may increase squat depth to three-quarters.

Suggestions

Increase squat depth; increase jogging speed and time.

Jill Michaels, CSCS

October 23, 2025

FIGURE 23.2 A sample strength and conditioning summary form, which records the athlete's current activities and responses to those activities.

Tissue Healing

The process of returning to competition following injury involves the healing of the injured tissues and the preparation of these tissues for the return to function. To better understand the role of the strength and conditioning professional during rehabilitation and reconditioning, it is necessary to review the general phases of tissue healing following musculoskeletal injury (34, 36). The timing of the events occurring within each phase of tissue healing differs for each tissue type and is affected by a variety of systemic and local factors, including age, lifestyle, degree of injury, comorbidities, and the structure that has been damaged. Moreover, the phases of healing progress along a continuum with no definitive timelines. However, all tissues follow the same basic pattern of healing ([table 23.1](#)).

TABLE 23.1 Tissue Healing

Inflammatory response phase ↓	Pain, swelling, and redness Decreased collagen synthesis Increased number of inflammatory cells
Fibroblastic repair phase ↓	Collagen fiber production Decreased collagen fiber organization Decreased number of inflammatory cells
Maturation—remodeling phase	Proper collagen fiber alignment Increased tissue strength

The process of returning to competition following injury involves healing of the injured tissues, preparation of these tissues for the return to function, and use of proper techniques to maximize rehabilitation and reconditioning.

Inflammatory Response Phase

Inflammation is the initial reaction to injury (34, 36) and is necessary for normal tissue healing to occur. Both local and systemic inflammation occur during the **inflammatory response** phase, allowing the eventual healing and replacement of damaged tissue. During the inflammatory phase, several events transpire that contribute to both tissue healing and an initial decrease in function. The injured area becomes red and swollen due to changes in vascularity, blood flow, and capillary permeability. After tissues are damaged, a locally hypoxic environment leads to a certain amount of tissue death that allows the release of several chemical mediators, including histamine and bradykinin. These substances further increase blood flow and capillary permeability in this local area, thereby promoting edema and fluid accumulation into the surrounding tissues. **Edema** elevates intraarticular pressure, which inhibits the motor neurons required for muscle activation (22, 84). Thus, edema can lead to muscle weakness and atrophy as well as limited function (22, 59). Tissue debris and pathogens (i.e., infectious agents) are removed from the injured area by increased blood flow and a process called *phagocytosis*; phagocytosis involves macrophages (i.e., a type of white blood cell), which engulf cellular debris to speed tissue healing. Additionally, fibroblast cells in nearby tissues are activated and begin to migrate toward the injury site to assist with healing. At this time, macrophages and platelets within the injured area release **growth factors** that assist in the transition to the fibroblastic repair phase (see next section).

The inflammatory substances present during this phase may result in stimulation of pain receptors (i.e., nociceptors), causing the injured athlete to sense pain, further contributing to decreased function. This phase typically lasts 72 hours following an acute injury but may last longer with a compromised blood supply and more severe structural damage. Though the inflammatory phase is critical to tissue healing, if it does not end within a reasonable amount of

time, the phases that follow may not occur, thereby delaying the rehabilitation process.

Fibroblastic Repair Phase

Once the inflammatory phase has subsided, tissue **repair** begins (34, 36); the **fibroblastic repair** phase, also known as the *proliferative phase*, is characterized by catabolism (tissue breakdown) and replacement of tissues that are no longer viable following injury. In an attempt to improve tissue integrity, new capillaries and connective tissue (scar tissue) form in the area. Fibroblasts from the inflammatory phase, with the help of growth factors, begin to synthesize type III collagen, which is randomly deposited along the injured structure and serves as the framework for tissue regeneration. This newly formed tissue is weaker than the original tissue; thus optimal strength of the new tissue is not yet achieved. Collagen fibers are strongest when they lie longitudinally (i.e., parallel) to the primary line of stress, yet many of the new fibers are positioned transversely, which limits their ability to efficiently transmit force. This phase of tissue healing begins as early as 72 hours after injury and may last up to two months.

Maturation–Remodeling Phase

The maturation–remodeling phase of healing often overlaps with the fibroblastic reparative phase and may begin as early as a few weeks after injury. During the **maturation–remodeling** phase of healing the weakened tissue produced during the repair phase is strengthened (34, 36). During this phase, new collagen fibers have shifted to a stronger type I collagen, allowing the newly formed tissue the opportunity to improve its structure, strength, and function. With increased loading, the collagen fibers of the newly formed scar tissue (a collection of cells and collagen that have repaired damaged tissue) begin to hypertrophy and align themselves along the lines of stress (29). The thicker and more optimally aligned collagen fibers become stronger, allowing a return to function.

Although strength of the collagen fibers and healing tissue improves, this tissue is not as strong as the normal healthy tissue it has replaced. Tissue **remodeling** in the maturation–remodeling phase can last months to years after injury (34, 36, 54).

Following injury, all damaged tissues go through the same general phases of healing: inflammation, repair, and remodeling. The timing of the events occurring within each phase of tissue healing differs for each tissue type and is affected by a variety of systemic and local factors, including age, lifestyle, degree of injury, comorbidities, and the structure that has been damaged. Characteristic events define each phase and separate one phase from another.

Goals of Rehabilitation and Reconditioning

The strength and conditioning professional must consider both the athlete's subjective response to injury and the physiological mechanisms of tissue healing; both are essential in relation to an athlete's return to optimal performance. The process of returning to competition following injury involves healing of the injured tissues, preparation of these tissues for the return to function, and use of proper techniques to maximize rehabilitation and reconditioning. While the goal is often a rapid resumption of activity, it is important to remember that each athlete responds differently to injury and thus progresses uniquely during rehabilitation.

As a preface to discussion of the goals of treatment during injury rehabilitation, two points must be made. First, healing tissue must not be overstressed (70). During tissue healing, controlled therapeutic stress is necessary to optimize collagen matrix formation (8, 29), but too much stress can damage new structures and significantly slow the athlete's return to competition. This means choosing a level of loading that provides neither too much stress nor too little stress to the athlete's healing tissue. It should be obvious that when one is choosing the load, it is necessary to consider the phase of healing and athlete type. For example, an exercise that provides too little tissue loading during the maturation–remodeling phase may provide too much loading (increased tissue stress) during the inflammatory response phase. Further, an exercise that provides too little stress to a professional basketball center may be excessive for an amateur cross country runner. The plane of movement is another necessary consideration. As an example, the medial collateral ligament of the knee is subject to the greatest stress when loaded in the frontal plane (i.e., valgus stress) during terminal knee extension. Therefore, if the athlete has a medial collateral ligament injury, frontal plane movements (valgus stress) such as a lateral box

jump or a side (lateral) shuffle should be avoided during early healing phases. However, those frontal plane movements should probably be included in some form during the later phases.

Second, the athlete must meet specific objectives to progress from one phase of healing to the next (98, 99). These objectives may depend on range of motion, strength, or activity. It is the responsibility of the team physician, athletic trainer, physical therapist, or a combination of these professionals to establish these guidelines.

Healing tissue must not be overstressed, but controlled therapeutic stress is necessary to optimize collagen matrix formation. The athlete must meet specific objectives to progress from one phase of healing to the next.

Inflammatory Response Phase

The first response to an injury is inflammation, a reaction that is essential for subsequent healing but also important to manage properly so as to not retard the rehabilitation process.

Treatment Goal

The primary goal for treatment during the inflammatory response phase is to prevent disruption of new tissue. A healthy environment for new tissue regeneration and formation is essential for preventing prolonged inflammation and disruption of new blood vessel and collagen production, which can prolong the injury. Relative rest from strenuous activity and **physical agents** (e.g., modalities) including ice, compression, elevation, and electrical stimulation are often primary treatment options to minimize tissue damage and decrease acute pain, but results on efficacy have been mixed (63, 93). One important consideration is that evidence has shifted from the Protection Rest Ice Compression Elevation (PRICE) approach to the Protection Optimal Loading Compression Ice Elevation (POLICE)

approach for the management of injuries in the inflammatory stage of healing as a result of improved outcome measures and faster recovery (5, 20).

It is also important to realize that a quick return to function relies on the health of other body tissues. Therefore, the power, strength, and endurance of the musculoskeletal tissues and the function of the cardiorespiratory system must be maintained. The strength and conditioning professional can provide significant knowledge and expertise in this area. To accomplish these tasks, the strength and conditioning professional should consult with the athletic trainer to determine which types of exercises are indicated and contraindicated for the specific injury. Maximal protection of the injured structures is the primary goal during this phase. Assuming that this requirement is fulfilled, exercises may include general aerobic and anaerobic training and resistance training of the uninjured extremities. If movement of the injured limb is not contraindicated, isolated exercises that target areas proximal and distal to the injured area may also be permissible provided that they do not stress the injured area. Examples include hip abduction and external rotation exercises following a knee injury (24, 39, 65) or scapula-stabilizing exercises following a shoulder injury (57, 94, 100).

To Ice or Not to Ice

Ice applications, also referred to as **cryotherapy**, have been used for centuries to decrease pain and inflammation following an injury (51). Advancements in cryotherapy methods from ice packs or ice massage and cold-water immersion to more advanced devices such as continuous cooling systems, vapor and gaseous sprays, and total body cooling have emerged, providing numerous application options. As part of the POLICE method for the management of acute injuries, early pain-free movement and ice are emphasized (5, 20). However, the use of ice has been questioned in both the strength and conditioning and allied health professions due to the potential for delaying natural post-injury healing processes and impeding muscle performance (52). These concerns, largely unsupported in the literature, stem from the potential of ice to cause a vasoconstriction of blood vessels (45), which may limit the ability of healing agents to reach the site of injury. While this concern is reasonable, there is no evidence that icing alone

following an injury may impede healing in human subjects. Contradictory evidence exists to determine if reductions in muscle performance occur following cryotherapy (31, 41).

Evidence pertaining to cryotherapy after injury has supported a pain-relieving analgesic effect via reduced nerve conduction velocity, with additional benefits that include a suppression of the inflammatory response with the potential to reduce swelling and enhanced return to sport (6, 31, 35, 37, 51, 96). Following exercise, evidence supports the use of cryotherapy to reduce symptoms of delayed onset muscle soreness (DOMS); however, systemic reductions in insulin-like growth factor (IGF) and protein synthesis may occur, impairing recovery (23, 73, 96). Thus, cryotherapy is recommended in the acute inflammatory stage as part of the POLICE strategy (20). A 15-minute duration is adequate when ice packs, ice massage, and cold-water immersion are used (31). Cold-water immersion ($<59^{\circ}\text{F}$ [$<15^{\circ}\text{C}$]) is a more efficient option when a larger coverage area is needed (e.g., whole hand or ankle) and is likely to have a greater effect on cooling deeper tissue (31). Local cryotherapy applications beyond the acute inflammatory phase are not warranted. Although cryotherapy may reduce DOMS and be effective when rapid recovery is needed, routine use is not recommended because it may impair protein synthesis (23).

Exercise Strategies

Although a rapid return to competition is often a goal, protective loading, also known as *active rest*, of the injured area is initially necessary to safeguard the damaged tissue from additional injury. Therefore, painful or intense exercise directly involving or stressing the injured area is not recommended during this phase. Exercises that do not directly involve or stress the injured area (e.g., upper extremity exercises for lower extremity injury, single-leg exercises with the uninvolved limb) and pain-free active movement of the involved area of injury may still be performed.

Fibroblastic Repair Phase

After the inflammatory response phase, the body begins to repair the damaged tissue with similar new tissue; however, the resiliency of this new tissue is low at this time. Repair of the weakened injury site can take up to eight weeks if the proper amount of restorative stress is applied, or longer if too much or too little stress is applied.

Treatment Goal

The treatment goal during the fibroblastic repair phase is to prevent excessive muscle atrophy and joint deterioration of the injured area. In addition, a precarious balance must be maintained in which disruption of the newly formed collagen fibers is avoided but low-load stresses are gradually introduced to allow increased collagen synthesis and prevent loss of joint motion. To protect the new, relatively weak collagen fibers in the early reparative phase, the athlete should avoid active resistive exercise involving the damaged tissue and may opt for submaximal activities such as functional movements against gravity or isometric resistance. Too little activity, though, can also have a deleterious effect, because newly formed fibers will not optimally align and may form adhesions (abnormal collection of cells and collagen that lack pliability), thereby preventing full motion. Early protected motion hastens the optimal alignment of collagen fibers and promotes improved tissue mobility. As in the inflammatory phase, therapeutic modalities are permissible, but their goal during repair is to promote collagen synthesis and manage pain. Ultrasound, electrical stimulation, and ice are continued in order to support and hasten new tissue formation (40, 82). Again, maintenance of muscular and cardiorespiratory function remains essential for the uninjured areas of the body. The strength and conditioning professional has considerable expertise to offer the other members of the allied health team regarding selection of the appropriate activities. Possible exercise forms during the repair phase include strengthening of the uninjured extremities and areas proximal and distal to the injury, aerobic and anaerobic exercise, and improving strength and neuromuscular control of the involved areas.

Exercise Strategies

The following exercises should be used during the repair phase only after consultation with the team physician, athletic trainer, or physical therapist. Isometric exercise may be performed, provided that it is pain free and otherwise indicated by the physician or

athletic trainer. Submaximal isometric exercise allows the athlete to maintain neuromuscular function and improve strength with movements performed at an intensity low enough that the newly formed collagen fibers are not disrupted. Unfortunately, isometric strengthening is joint angle specific; that is, strength gains occur only at the angles used (46). Therefore, if indicated, it may be appropriate for the athlete to perform isometric exercises at multiple angles (46). Isokinetic exercise uses equipment that provides resistance to movement at a given speed (e.g., $60^{\circ}/s$ or $120^{\circ}/s$). Because sport is not performed at a single consistent speed, isokinetic exercise may be limited in real-world application as a standalone exercise. Furthermore, most isokinetic equipment allows single-joint exercise only, which permits concentration on a specific muscle or joint but is not always the most functional method of strengthening. While isokinetic machines may have a more limited application in terms of strengthening specificity, these devices may be of value for providing continuous passive motion early in the post-injury or post-operative phases of healing and can be used to assess muscle balance or ratios (e.g., quadriceps-to-hamstring strength ratio).

While isotonic exercise (i.e., concentric and eccentric) involves movements with constant external resistance, the amount of force required to move the resistance varies, depending primarily on joint angle and the length of each agonist muscle. Isotonic exercise uses several different forms of resistance, including gravity (i.e., exercises performed without equipment, with gravitational effects as the only source of resistance), dumbbells, barbells, and weight stack machines. Concentric and eccentric muscle actions can be used to increase strength and appropriately stress healing tissues. Eccentric exercise allows for greater force production and requires less energy expenditure compared to concentric exercise and has been shown to promote healing or tissue repair via the concept of **mechanotransduction** (44, 61). Mechanotransduction occurs when tissues are subject to overload, which communicates a

mechanical message from the extracellular to intracellular region of the tissue. This mechanical message upregulates the cellular DNA, which in turn stimulates collagen synthesis (44). Loads may be increased to provide greater challenge as tissue healing progresses. The speed at which the movement occurs is controlled by the athlete; movement speed can be a program design variable, with more acute injuries calling for slower movement and the later phases of healing better suited to faster, more sport-specific movement.

Neuromuscular control is the ability of muscles to respond to sensory feedback to maintain joint stability (85). This sensory information is referred to as **proprioception** and occurs in response to receptors in the skin, muscles, tendons, ligaments, and the joint capsule. Proprioception contributes to the conscious and unconscious efferent control of posture, balance, stability, and sense of position (85). For example, when running on an uneven surface, cross country runners rely on sensory input (proprioception) from their lower extremities to adjust to the ground to prevent falls and injuries. After an injury, neuromuscular control is usually impaired due to **arthrogenic inhibition**, which occurs from swelling within the joint that causes a reflex inhibition of the alpha motor neuron responsible for muscle activation (22). Specific types of exercises exist to improve neuromuscular control following injury and can be manipulated through alterations in surface stability, vision, and speed. Mini-trampolines, balance boards, and stability balls can be used to create unstable surfaces for upper and lower extremity training. Athletes can perform common activities such as squats and push-ups on uneven surfaces to improve neuromuscular control. Exercises may also be performed with eyes closed, thus removing visual input, to further challenge balance. Finally, increasing the speed at which exercises are performed provides additional challenges to the system. Specifically controlling these variables within a controlled environment that is sport or activity specific will

allow the athlete to progress to more challenging exercises in the next stage of healing.

Maturation–Remodeling Phase

The outcome of the repair phase is the replacement of damaged tissue with collagen fibers. After those fibers are laid down, the body can begin to remodel and strengthen the new tissue, allowing the athlete to gradually return to full activity.

Treatment Goal

The primary goal during the maturation–remodeling phase is to optimize tissue function while transitioning to return to play or activity. Athletes improve function by continuing and progressing the exercises performed during the repair phase and by adding more advanced, sport-specific exercises that allow progressive stresses to be applied to the injured tissue. The athlete can be tempted to do too much too soon, which may further damage the injured tissues. It is important to remember that, while there may be less pain with activity at this point, the injured tissues have not fully healed and require further attention to achieve complete recovery ([figure 23.3](#)). Progressive tissue loading allows improved collagen fiber alignment and fiber hypertrophy (8, 30). Return-to-play or activity decisions should be based on an understanding of normal tissue healing time frames as well as a criteria-based progression with predetermined objectives. These objectives usually include measures of range of motion and strength, functional testing, and patient-reported function using established patient-reported outcome forms (1, 57, 100). It is important for the allied health team members to communicate, have clearly defined roles, and use return-to-play metrics to ensure a safe return to play.

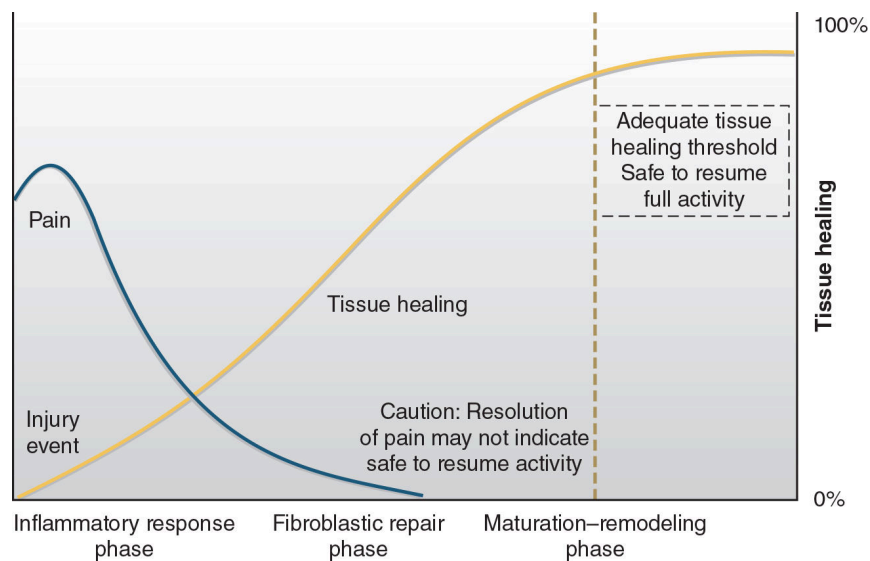


FIGURE 23.3 Profile of a typical soft-tissue injury response. Pain is often used as a guide for tissue health. Pain levels (blue line) often decrease well before tissue healing (yellow line) is complete, which may lead athletes to inappropriately believe adequate tissue healing has occurred (vertical yellow line) and it is safe to resume full activity.

Exercise Strategies

Ultimately, as with other aspects of strength and conditioning, rehabilitation and reconditioning exercises must be functional and must mimic activity demands—that is, reflect specificity. Examples of functionally specific training include joint angle-specific strengthening, velocity-specific muscle activity, closed kinetic chain exercises, and exercises designed to further enhance neuromuscular control. Strengthening should transition from general exercises to sport-specific exercises designed to replicate movements common in given sports. For example, for a basketball guard who sprained an ankle, rehabilitation exercises may progress from a general single-joint exercise to those more specific to the sport and position ([figure 23.4](#)). Specificity of movement speed is another important program design variable. Strengthening exercises performed during rehabilitation should also mimic sport speed requirements; that is, activities that require greater speeds (e.g., sprinting) should include exercises performed at higher velocities. Using the example in [figure 23.4](#), the exercises progress from the relatively slow and controlled balance and strength exercises to the faster plyometric and sprinting

exercises. Similarly, consider a sprinter with a hamstring muscle strain. Whereas initial reconditioning may concentrate on the recovery of flexibility and strength of the injured muscles, the nature of this athlete's sport necessitates exercises performed at rapid speeds during the later phases of rehabilitation and reconditioning. Exercise selection for a sprinter with an improving hamstring muscle strain might progress from hamstring flexibility to eccentric strength to concentric strength to dynamic stretching and finally to rapid isotonic strengthening. Examples of velocity-specific exercise include manually resisted strengthening and plyometric and speed training. Refer to [chapters 19](#) and [20](#) for a thorough discussion of plyometric and speed training, respectively.



FIGURE 23.4 Sample progression for a basketball player with a goal of returning from an ankle sprain. Exercises progress from general to basketball specific.

The kinetic chain is the collective effort or involvement of two or more sequential joints to create movement (88). A **closed kinetic chain** exercise is one in which the terminal joint meets with considerable resistance that prohibits or restrains its free motion (88); that is, the distal joint segment is stationary (e.g., fixed to the

ground). Lower extremity closed kinetic chain exercises have often been classified as a more functional form of exercise than open kinetic chain exercises (2, 94) because most sport-related activities are performed with the feet “fixed” to the playing surface. For example, during the closed kinetic chain squat exercise, the feet are “fixed” to the floor and essentially do not move, providing a base on which movement occurs ([figure 23.5a](#)). Closed kinetic chain exercises have several advantages, including increased joint stability and functional movement patterns; during sport activity, joints are not typically used in isolation but rather work in concert with the adjacent joints and surrounding musculature. Although closed kinetic chain exercises are commonly viewed as lower extremity exercises, closed chain upper extremity exercises exist as well ([figure 23.5b](#)) (57, 100).



FIGURE 23.5 Closed kinetic chain exercises: (a) squat exercise and (b) push-up exercise.

An **open kinetic chain** exercise uses a combination of successively arranged joints in which the terminal joint is free to move; open kinetic chain exercises allow for greater concentration on an isolated joint or muscle (2). An example is the leg (knee) extension exercise, during which the feet and lower legs are allowed to move freely ([figure 23.6](#)). The leg extension allows greater concentration on the quadriceps at the knee joint; in comparison, the squat, which also uses the quadriceps muscles and knee joint, relies on muscle activity on the hip and ankle joints as well. Although closed kinetic chain exercises are often viewed as more functional,

most activities involve both closed and open kinetic chain movements. In sprinting, for example, while one lower extremity is on the ground (closed kinetic chain), the other is in the air (open kinetic chain), which means that both types of movement can occur simultaneously. In some situations, an open chain exercise may therefore be an equally appropriate choice. Some examples of upper extremity open kinetic chain activities include the biceps curl and bench press exercises.



FIGURE 23.6 Example of an open kinetic chain exercise—leg (knee) extension exercise.

Exercises designed to improve neuromuscular control, which were introduced during the repair phase, should be continued and progressed as appropriate during the maturation–remodeling phase. Rehabilitation and reconditioning goals and strategies are summarized in the sidebar.

Rehabilitation and Reconditioning Goals and Strategies

Although rehabilitation and reconditioning programs must be individualized, the following are lists of general goals and approaches for each phase.

Inflammatory Response Phase

- Prevention of new tissue disruption and prolonged inflammation with the use of relative rest and passive modalities
- Maintenance of function of the cardiorespiratory and surrounding neuromusculoskeletal systems
- No active exercise for the injured area

Fibroblastic Repair Phase

- Prevention of excessive muscle atrophy and joint deterioration of the injured area
- Maintenance of function of the neuromusculoskeletal and cardiorespiratory systems
- Possible exercise options: submaximal isometric, isokinetic, and isotonic exercise; balance and proprioceptive training activities

Maturation–Remodeling Phase

- Optimization of tissue function
- Progressive loading of the neuromusculoskeletal and cardiorespiratory systems as indicated
- Possible exercise options: joint angle–specific strengthening, velocity-specific muscle activity, closed and open kinetic chain exercises, proprioceptive training activities

Program Design

The area in which strength and conditioning professionals can best contribute to the rehabilitation and reconditioning process is the provision of resistance and aerobic training programs designed for the injured athlete. Their experience in prescribing exercise for uninjured athletes gives strength and conditioning professionals the ability to properly adapt training programs for those recovering from injury. Although protocols do exist for exercise prescription following injury, many do not incorporate sport-specific program design variables; the same principles used to design resistance and aerobic training programs for uninjured athletes should be applied during rehabilitation and reconditioning.

Resistance Training

Several programs have been developed to assist with the design of resistance training programs (16, 21, 47, 89, 104), and many of these programs have been advocated for use in the rehabilitation setting (16, 47, 104). Both the DeLorme (16) and Oxford (104) programs use 3 sets of 10 repetitions with a pyramid-type design. DeLorme's program progresses from light to heavy resistance. The initial set involves 10 repetitions of 50% of the athlete's 10-repetition maximum (10RM), the second set increases the resistance to 75% of the 10RM, and the final set requires 100% of the 10RM (16, 17). The Oxford system (104) is the reverse of DeLorme's system; that is, it progresses from heavy to light resistance. The first set performed is 100% of the athlete's 10RM, the second is 75% of the 10RM, and the third is 50% of the 10RM (104). Unlike the DeLorme system, which has a built-in warm-up from progressive increases in resistance, the Oxford system begins with heavy resistance. Thus, a non-fatiguing warm-up set should be considered.

Knight's (47) **daily adjustable progressive resistive exercise (DAPRE)** system requires and allows more manipulation of intensity and volume than either the DeLorme or the Oxford system (16, 17,

104); DAPRE involves 4 sets, with repetitions ranging from 10 to possibly only 1 during the final set. The first set requires 10 repetitions of 50% of the estimated 6RM, and the second set requires 6 repetitions of 75% of the estimated 6RM. The third set in the DAPRE requires the maximum number of repetitions of 100% of the estimated 6RM; the number of repetitions performed during the third set determines the adjustment made in resistance for the fourth set ([table 23.2](#)) (48). The DeLorme, Oxford, and Knight systems are protocols that have been shown to increase muscle strength (17, 48, 104) and may be appropriate when one is designing rehabilitation resistance training programs. However, athletes require rehabilitation and reconditioning programs tailored to the demands of their respective sports. Although these protocols (17, 48, 104) are established strengthening programs, they may be too strict to allow individualization for athletes in different sports.

TABLE 23.2 Resistance Adjustments for Daily Adjustable Progressive Resistance Exercise (DAPRE) Protocol

Number of repetitions performed during set 3	Adjusted resistance for set 4	Resistance for next exercise session
0-2	Subtract 5-10 lb (2.3-4.5 kg) and repeat the set	—
3-4	Same resistance or subtract 5 lb (2.3 kg)	Same resistance
5-7	Same resistance	Add 5-10 lb (2.3-4.5 kg)
8-12	Add 5-10 lb (2.3-4.5 kg)	Add 5-15 lb (2.3-6.8 kg)
13+	Add 10-15 lb (4.5-6.8 kg)	Add 10-20 lb (4.5-9.1 kg)

Adapted by permission from K. L. Knight, "Knee Rehabilitation by the Daily Adjustable Progressive Resistive Exercise Technique," *American Journal of Sports Medicine* 7, no. 6 (1979): 336-337, Copyright © 1979. Reprinted by Permission of Sage Publications.

Programs for both the healthy and the damaged tissues of an injured athlete require the same basic design principles provided in [chapter 18](#). According to the specific adaptation to imposed demands (SAID) principle, the system will adapt to the demands placed on it. Therefore, the goal of training (specific adaptation) should dictate the design of the resistance training program (imposed demands). For example, during the remodeling phase of an injured marathoner's patellofemoral rehabilitation, his or her quadriceps muscles should be trained with an emphasis on muscular endurance. Therefore, the athlete should perform many repetitions of the rehabilitation exercises to prepare the muscles for the demands of long-distance running. Conversely, a weightlifting athlete rehabilitating from the same injury requires fewer repetitions of high-intensity rehabilitation and reconditioning exercises during the later portions of the remodeling phase to prepare the muscles for the power demands of this sport. Refer to [table 23.3](#) for a comparative example of the rehabilitation strategies for these two athletes.

Aerobic and Anaerobic Training

Although research has yet to determine an optimal aerobic training program for use in the rehabilitation setting, it is generally accepted that the program should mimic specific sport and metabolic demands as closely as possible. The strength and conditioning professional, having a background in designing and implementing conditioning exercises for healthy athletes, is the ideal member of the allied health team to prescribe and supervise the aerobic training portion of the injured athlete's reconditioning program. As with the resistance training program, the strength and conditioning professional must consider the demands that the given sport places on the injured athlete. Keeping these demands and the contraindications of the injury in mind, the strength and conditioning professional can use the prescription guidelines in [chapters 17](#) and

[20](#) to create an appropriate training program to allow an uncomplicated return to competition.

TABLE 23.3 Application of Design Principles for Resistance Training Programs During Rehabilitation and Reconditioning From Patellofemoral Injury

Phase of healing	Design variable	Athlete	
		Marathon runner	Weightlifter
Inflammatory response phase	Goals and exercises	• None that require quadriceps activity—relative rest of this area needed to reduce inflammation • Maintenance of muscular strength and endurance of adjacent areas (i.e., hip extensors, knee flexors, and plantar flexors) • Maintenance of cardiovascular fitness	• None that require quadriceps activity—relative rest of this area needed to reduce inflammation • Maintenance of muscular strength and power of adjacent areas (i.e., hip extensors, knee flexors, and plantar flexors) • Maintenance of upper body muscular strength and power
Fibroblastic repair phase	Goals and exercises	• Isometric quadriceps strengthening at full knee extension (progressing to multiple angles) • Progress to pain-free isotonic quadriceps strengthening (following consultation with sport medicine team) • Continue exercise of adjacent areas • Continue aerobic exercise; may begin stationary bicycle or stair stepper (per recommendation of sports medicine team)	• Isometric quadriceps strengthening at full knee extension (progressing to multiple angles) • Progress to pain-free isotonic quadriceps strengthening (following consultation with sport medicine team) • Continue exercise of adjacent areas • Continue upper body muscular strength and power exercises
	Sets × repetitions	2-3 × 15-20	3-4 × 8-10

	Intensity	Submaximal ($\leq 50\%$ 1RM)	Submaximal ($\leq 50\%$ 1RM)
Maturation–remodeling phase	Goals and exercises	• Begin more sport-specific activities, movements, and speeds • Return to running, gradually increasing distance and speed as tolerated • Add lunge and squat (increase knee range of motion as able)	• Begin more sport-specific activities, movements, and speeds • Increase movement speeds to those similar to competition • Add Romanian deadlift and squat (increase knee range of motion as able)
	Sets $\times$ repetitions	2-3 $\times$ 15-20	4-5 $\times$ 3-8
	Intensity	Progress to maximal intensity (50%-75% 1RM)	Progress to maximal intensity ($>75\%$ 1RM)

Again consider rehabilitation from a patellofemoral injury for a marathon runner, a wrestler, and a weightlifter; the metabolic energy demands of their rehabilitation and reconditioning are markedly different. For the marathon runner's goals, aerobic fitness is of much greater concern and must be addressed immediately. The metabolic demands for the wrestler involve a combination of aerobic and anaerobic systems; therefore, interval training is more appropriate (27). In contrast, the weightlifter's program would focus on maintaining anaerobic fitness. Selecting the appropriate training device depends on the region of the body that is injured. While specificity of training is important, gains in exercise capacity can be made through exercise of other body regions (i.e., upper extremity exercise to improve oxygen consumption) (92). A variety of options exist to modify aerobic and anaerobic training, including the upper body ergometer, deep-water running, lower extremity cycling, and elliptical machines. Strategies to maintain cardiovascular fitness can be implemented early, even during the inflammatory response phase. As already emphasized, it is important to remember that to allow optimal healing, attempts should be made to initially minimize stress to healing tissues, but relevant stresses can be applied to

uninjured areas. An example would be for a soccer player with an acute right knee injury. Although options for right lower extremity strengthening may be initially limited, exercises may still be performed for the upper extremity as well as the uninvolved left lower extremity (e.g., single-leg squat with weights). The strength and conditioning professional may need to modify uninvolved left lower extremity exercises to ensure that the involved right lower extremity is protected. Exercises targeting the uninvolved limb can improve muscle strength in the involved limb (55, 56). Thus it is important for injured athletes to continue to perform strength and conditioning exercises for the uninvolved body regions.

Designing strength and conditioning programs for injured athletes requires the strength and conditioning professional to examine the rehabilitation and reconditioning goals to determine what type of program will allow the quickest return to competition.

Reducing Risk of Injury and Reinjury

In addition to using strength training and conditioning strategies when rehabilitating athletes from injuries, strength and conditioning professionals can implement research findings to reduce both risk for initial injury and risk of reinjury following injury (77, 90). Structured programs have been developed for the lower (25, 28, 33, 76) and upper extremities (57, 101). These programs are often sport specific and address common risk factors for injury.

Previous injury is one of the most substantial risk factors for future injury in active individuals (30, 75, 86, 91). Risk factors for upper extremity injury include decreased glenohumeral range of motion, scapular dyskinesis (i.e., abnormal scapular movement), and decreased shoulder strength (9, 13, 101). Among overhead athletes, shoulder internal rotation range of motion restrictions have been shown to increase the risk for shoulder injury, whereas shoulder external rotation and flexion range of motion restrictions are associated with elbow injuries (7, 11, 42). Range of motion exercises and the Throwers Ten are often used as a structured program to reduce upper extremity injury risk (102). The Throwers Ten program includes a specific set of upper extremity exercises and movements designed to improve muscle performance as required for throwing activities (26). Additional exercises shown to reduce the risk of upper extremity injury among athletes include strengthening the shoulder external rotators ([figures 23.7a, b](#)) and stretching the shoulder to improve internal rotation ([figures 23.8a, b](#)) (14, 49, 64, 66, 78). Risk factors for lower extremity injury include decreased balance, decreased neuromuscular control during jump landing, and decreased lower extremity muscle strength (3, 53, 103). Structured lower extremity programs to reduce injury risk should be sport specific and should focus on neuromuscular control during activities such as landing from a jump and cutting. Two exercises that may be used to reduce this risk of injury to the lower extremities are proper jumping and landing technique during plyometric exercise ([figure](#)

[23.9a](#)) and a single-leg squat to emphasize unilateral strength ([figure 23.9b, c](#)).

Previous injury is one of the most substantial risk factors for future injury in active individuals. Lower extremity programs designed to reduce injury risk should be specific to a sport's demands and should focus on the neuromuscular control required during specific activities such as landing from a jump and cutting.

While upper extremity injury risk-reduction programs exist (102), much of the research has focused on the prevention of lower extremity injuries, specifically anterior cruciate ligament (ACL) injuries. Two examples of structured programs designed to reduce injury risk are Sportsmetrics (33) and PEP (Prevent Injury and Enhance Performance) (62). While using slightly different foci, each of these programs introduces exercises and movements shown to address risk factors for injury and reduce lower extremity injury rates, especially ACL injuries and ankle sprains (25, 28, 33, 76). In addition, using eccentric exercise has been shown to dramatically reduce the risk of hamstring injury (74, 79).

After return to sport, individuals may continue to demonstrate deficits in strength, biomechanics, and functional performance (71, 87). Deficits may often be compared to the contralateral limb (60) or established normative values (19, 38). Side-to-side differences of less than 10% in strength and functional performance may be considered acceptable (60). The transition between supervised rehabilitation for an injury and unrestricted activity participation is a critical time when members of the allied health team need to be in communication regarding how deficits in strength and performance will be managed.

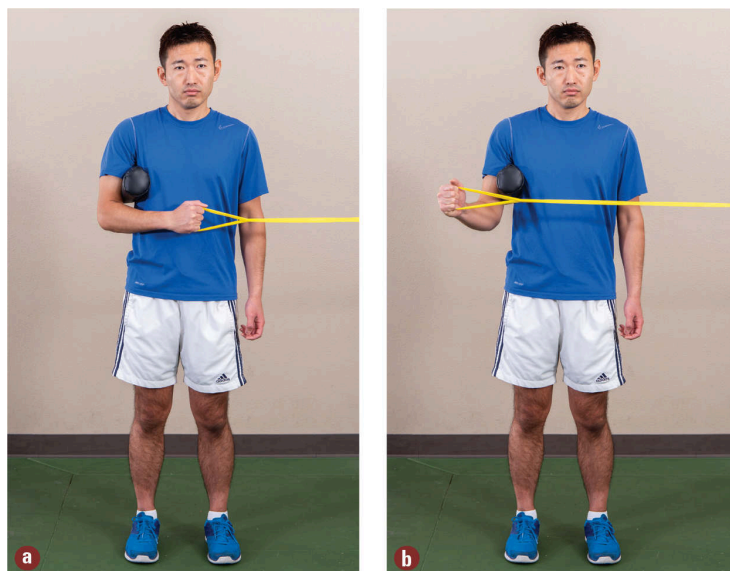


FIGURE 23.7 Shoulder external rotator strengthening may be used to reduce shoulder injuries and improve shoulder function and stability: (a) start position and (b) end position.

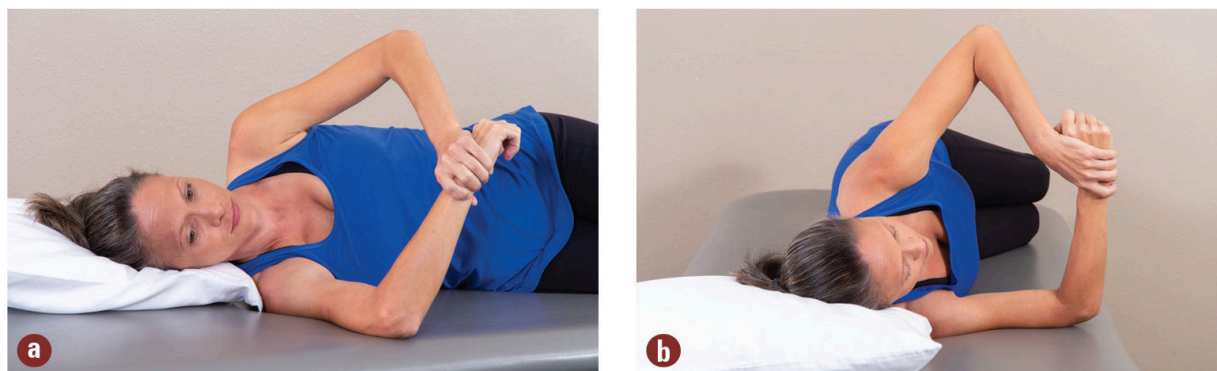


FIGURE 23.8 The sleeper stretch is used to reduce shoulder injuries, improve shoulder internal rotation, and reduce posterior shoulder tightness; (a) front view and (b) top view.

Medical Conditions

Strength and conditioning professionals must have an understanding of medical conditions that may develop from training as well as preexisting conditions that may increase the risk of an adverse event from exercise. Possessing this knowledge is essential for ensuring safe participation in exercise and appropriate discontinuation procedures. Recognizing the early presentation and management strategies for rhabdomyolysis, sickling from the sickle cell trait, and temperature-induced illnesses are a necessary requirement for training both the athletic and general populations.

Rhabdomyolysis

Rhabdomyolysis is characterized by the breakdown of skeletal muscle tissue, leading to the release of myoglobin into the bloodstream. Myoglobin can pose a risk to kidney function, potentially causing kidney damage or failure if left untreated. Various factors can trigger rhabdomyolysis, including trauma, extreme physical exertion, and muscle compression. The severity of rhabdomyolysis ranges from mild muscle discomfort to severe, life-threatening complications (43). Symptoms of rhabdomyolysis may manifest as severe muscle pain and weakness, dark urine (often resembling cola, properly termed *myoglobinuria*), swelling, tenderness of affected muscles, alongside fatigue, nausea, vomiting, and confusion. In severe instances, individuals may experience dehydration, electrolyte imbalances, and acute kidney injury (43).

If rhabdomyolysis is suspected, immediately discontinue activity, lie the client down, implement proper hydration protocols, and closely monitor to ensure stability while awaiting further medical assistance (10). Prompt referral to a medical professional is necessary to prevent more serious medical complications (10).



FIGURE 23.9 Two exercises that may be used to reduce risk of injury to the lower extremities are (a) proper jumping and landing technique during plyometric exercise and (b, c) a single-leg squat to emphasize unilateral strength.

Prevention of rhabdomyolysis is best achieved by proper exercise prescription. Avoidance of excessive or unaccustomed exercise intensity, volume, and duration as well as consideration of baseline fitness levels before engaging in new exercise serves as a key preventative measure.

Sickle Cell Trait

The **sickle cell trait** (SCT) is an inherited condition characterized by one abnormal hemoglobin gene and one normal hemoglobin gene. Individuals with SCT may experience complications akin to sickle cell disease under specific circumstances like extreme physical exertion or low oxygen levels (72). Unlike **sickle cell disease**, which necessitates inheriting two abnormal genes, SCT carriers usually do not exhibit severe symptoms. In situations such as high-altitude training environments or extreme exertion where oxygen levels are low, the red blood cells can change to an abnormal shape (sickling) and begin to clump together, which ultimately may be responsible for constriction of the blood vessels. When sickling and the associated vascular occlusion occurs, reports of fatigue or tiredness, pain, swelling of hand and feet, and vision problems may occur (69). Longer term consequences of sickling include damage to the blood vessels; a shorter life span of existing red blood cells, leading to anemia; and, in cases of vascular occlusion, stroke. Evidence suggests that the SCT is associated with sudden death during periods of extreme exertion and is more common among young African Americans (69). A majority of deaths attributed to the SCT have occurred during military training and athletic participation at the collegiate level (69).

Should an individual develop symptoms of sickling, they should immediately discontinue physical activity and be removed from extreme temperature environments. Owing to potential dehydration, proper hydration protocols should be implemented immediately (95). Immediate referral to a medical professional is indicated if acute symptoms are present.

Key prevention strategies for SCT involve awareness of triggers for sickling, education, preventive measures, and regular medical follow-up. Testing for SCT is recommended among athletes and those engaged in intense training because SCT has been associated with sudden death in individuals during extreme exertion. Intense physical activity should be approached with caution, especially in extreme conditions or high altitudes, to reduce the potential for sickling (81). For individuals with the SCT who engage in exercise training, proper climate acclimatization, training intensity considerations to stay at or below the anaerobic threshold, and an active cool-down to promote lactate clearance will serve as a reasonable preventative strategy (68, 80).

Heat Cramps

Heat cramps are part of a continuum of heat-related injuries or illnesses marked by dysregulation of the body's thermoregulation capacity (58). **Heat cramps** are muscle spasms or involuntary sustained contractions of large muscle groups that may occur when the athlete is exposed to elevated environmental temperatures during exercise. Heat cramps may occur during or shortly after exercise. Several factors may contribute to heat cramps, including dehydration, depleted electrolytes, increased training load or duration, and muscular fatigue. Although heat cramps generally occur in hot environments, elevated core body temperature is not always present.

When heat cramps are suspected, immediate cessation of activity and moving the athlete to a cooler environment to lower body temperature is a necessity. Implementing proper rehydration is essential as are appropriate cooling measures. Passive gentle stretching or massage to affected muscle groups may be beneficial (58). Heat cramps may be prevented by proper hydration protocols, use of light-weight clothing when exercising in hot environments, and appropriate progressive conditioning in the specific training environment to allow heat acclimation (58).

Heat Syncope

Heat syncope occurs when an athlete experiences dizziness or lightheadedness and faints during or immediately after training (12). Heat syncope may occur during positional changes or with prolonged positioning during exercise in hot environments. Risk factors for heat syncope include training in hot environments without prior heat acclimation, dehydration, pooling of blood in the venous system, and low blood pressure (58). When syncope is suspected or fainting occurs it is important to lay the athlete in a supine position and elevate the legs above the level of the heart. If fainting has occurred, monitoring of vital signs and shielding of the body from direct heat exposure is recommended. Once appropriate, proper oral rehydration is necessary. Prevention strategies include proper acclimation to training in hot environments and ensuring proper hydration before, during, and after training. If low blood pressure or frequent syncope occurs despite preventative efforts, medical consultation is necessary.

Heat Exhaustion

Heat exhaustion is the inability to effectively exercise in the heat. Numerous factors contribute to heat exhaustion including low blood pressure, dehydration, lack of acclimation to exercise in hot environments, heavy sweating, and electrolyte depletion (12, 58). Symptoms of heat exhaustion include inability to continue exercise, pallor, heat cramps, weakness, syncope, dizziness, headache, hyperventilation, nausea, diarrhea, acute loss of appetite, decreased or dark urine output, and elevated core temperature up to 104.9 °F (40.5 °C) (12, 58).

If heat exhaustion is recognized, immediately remove the athlete from the hot environment and, once appropriate, restore oral rehydration. If oral rehydration is not possible due to nausea, parenteral (non-oral method such as intravenous) rehydration may be considered (58).

Aggressive cooling measures should be implemented such as forced air conduction or spraying of lukewarm water to cool the body (58). In cases where these measures are not effective, total body cooling in an ice bath may be indicated (12). It is important to immediately respond to symptoms of heat exhaustion to avoid progression to heatstroke.

Prevention strategies include proper acclimation to training in hot environments and ensuring proper hydration before, during, and after training.

Heatstroke

Heatstroke, a progression of heat exhaustion, is considered a medical emergency (12, 58). Heatstroke occurs from intense or long-term exercise and is characterized by body temperature dysregulation greater than heat exhaustion. Signs and symptoms of heatstroke include rapid heart rate, hypotension, sweating with possible evaporation of sweat by time of incident, hyperventilation, altered mental state, diarrhea, seizure, and coma (12, 58).

Immediate recognition of heatstroke is vital to preventing lethal consequences; core body temperatures in excess of 109.4 °F (43 °C) have been associated with death. Immediate total body cooling measures must be implemented such as total body immersion in ice. Monitoring of core body temperature is paramount to appropriately responding to heatstroke. Contacting emergency medical services is indicated in cases where medical staffing is not present.

Prevention strategies include proper acclimation to training in hot environments and ensuring proper hydration before, during, and after training. A key element of prevention resides in recognizing that heat illness exists on a continuum of events all related to dysregulation of the body's thermoregulatory capacity. As such, heatstroke is likely to be preceded by heat cramps, syncope, and heat exhaustion. Recognizing the presentation of heat cramps or heat exhaustion before heatstroke is paramount to prevention.

Cold Exposure

Cold exposure, or **hypothermia**, occurs when the body's temperature drops below 95 °F (35 °C), causing the physiological systems to reduce functions and eventually shut down (18, 50). While mild hypothermia is manageable, more severe forms are a medical emergency. Hypothermia affects all organ systems, and early symptoms include shivering; pale, dry skin; and increased muscle tone (18). Increased heart rate, blood pressure, and respiration may also be present as the body attempts to promote thermogenesis (18). Risk factors for hypothermia include exercising in cold temperatures without adequate warming methods as well as exposure to rain or cold water that results in wet clothing, which may promote further heat loss.

Upon recognizing symptoms of cold exposure, immediate measures for warming the body are required through environmental temperature and use of clothing. Monitoring body temperature is a necessity once hypothermia is recognized; core temperature less than 89.6 °F (32 °C) may lead to cardiovascular failure (18).

It is important to recognize that although the body can acclimate to training in hot conditions, there is little to no acclimation to training in cold environments. Prevention strategies include recognizing extremes in temperature exposure and use of protective clothing to help create a warming effect while training.

Conclusion

Effective athletic injury rehabilitation and reconditioning require efficient communication between the members of the allied health team. Each member plays a distinct role in ensuring the injured athlete's return to function; although the job of each professional is different, the tasks of each team member complement those of the others on the team. Goals are established for each athlete, and therapeutic exercise programs must be designed, administered, and progressed according to the tissue healing sequence. The program must be individualized to effectively return the athlete to normal function and competitive athletics. Early rehabilitation for initial tissue protection and strengthening is more structured than later rehabilitation phases; the latter allows for progression to functional activities that are specific to the athlete's sport and position. Designing reconditioning programs for injured athletes necessitates a careful examination of the sport requirements and a thorough understanding of both the healing process and therapeutic exercise.

KEY TERMS

allied health
arthrogenic inhibition
athletic trainer
closed kinetic chain
contraindication
contusion
counselor
cryotherapy
daily adjustable progressive resistive exercise (DAPRE)
dislocation
edema
exercise physiologist
fibroblastic repair

growth factors
heat cramps
heat exhaustion
heatstroke
heat syncope
hypothermia
indication
inflammation
inflammatory response
macrotrauma
maturation–remodeling
mechanotransduction
microtrauma
neovascularization
neuromuscular control
nutritionist
open kinetic chain
physical agent
physical therapist
physiotherapist
proprioception
psychiatrist
psychologist
remodeling
repair
rhabdomyolysis
sickle cell disease
sickle cell trait (SCT)
sprain
strain
strength and conditioning professional
subluxation
team physician
tendinitis

tendinopathy

STUDY QUESTIONS

1. Following an acute ankle sprain, which of the following measures should be immediately implemented to enable prompt recovery?
 - a. POLICE
 - b. PRICE
 - c. avoidance of ice
 - d. avoidance of ankle movement
2. Which of the following management strategies is recommended when heatstroke occurs?
 - a. passive stretching of large muscle groups
 - b. total body ice bath immersion
 - c. supine positioning with the legs elevated above the heart
 - d. direct exposure to forced air conduction
3. Following injury, which of the following healing factors are responsible for releasing growth factors?
 - a. fibroblasts
 - b. platelets
 - c. histamine
 - d. bradykinin
4. Which of the following conditions presents with myalgia, weakness, and myoglobinuria?
 - a. sickle cell trait
 - b. heat cramps
 - c. rhabdomyolysis
 - d. tendinopathy
5. Which of the following strategies has been associated with enhancing collagen synthesis and preventing hamstring strains?
 - a. concentric focused power activities
 - b. isometric performance at a submaximal level

- c. passive stretching
- d. eccentric focused overload